Generalized Gaussian Binary Classifier

Algorithm Reference and Mathematical Derivations

Algorithm Reference Document

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1 Introduction

This document provides a complete mathematical description of the **Generalized Gaussian Binary Classifier** (`gg_classifier.py`), a tool designed for binary classification of multi-band remote sensing imagery. The classifier extends the standard multivariate Gaussian model by replacing the quadratic exponent with a power-exponential form, yielding heavier or lighter tails than the normal distribution depending on a tunable shape parameter β.

The system supports two operational modes:

- **–fit**: estimate model parameters from a labelled raster and binary training mask; optionally select the most discriminative band subset via greedy KL-divergence optimization;
- **–apply**: load a serialized model and produce probability and classification maps for a new raster, with strict band-name validation.

2 Notation

Symbol	Meaning
N	Number of pixels (observations)
B	Number of spectral bands (features)
$\boldsymbol{x} \in \mathbb{R}^B$	Feature vector for a single pixel
$y \in \{0, 1\}$	Binary class label (0 = negative, 1 = positive)
$\boldsymbol{\mu}_c \in \mathbb{R}^B$	Class-conditional mean vector, class c
$\boldsymbol{\Sigma}_c \in \mathbb{R}^{B \times B}$	Class-conditional covariance matrix, class c
$\boldsymbol{\Sigma}_c^{-1}$	Precision matrix (inverse covariance)
π_c	Class prior, $\pi_c = P(Y = c)$
$\beta > 0$	Shape parameter of the Generalized Gaussian
τ	Decision threshold on posterior probability

3 The Generalized Gaussian Distribution

3.1 Univariate Generalized Gaussian

The univariate Generalized Gaussian (also called power-exponential) distribution with location μ, scale σ, and shape $\beta > 0$ has density:

$$p(x \mid \mu, \sigma, \beta) = \frac{\beta}{2\,\sigma\,\Gamma(1/\beta)} \exp\left(- \left| \frac{x - \mu}{\sigma} \right|^{\beta} \right) \tag{1}$$

Special cases:

- $\beta = 1$: Laplace (double-exponential) distribution — heavy tails;
- $\beta = 2$: standard Gaussian distribution — medium tails;
- $\beta \to \infty$: uniform distribution on $[\mu - \sigma, \mu + \sigma]$ — no tails.

3.2 Multivariate Generalized Gaussian (Power-Exponential)

For a B-dimensional observation $\boldsymbol{x}$, the multivariate generalization replaces the squared Mahalanobis distance with its $\beta/2$-th power.

Definition 1 (Mahalanobis distance)**.** *The squared Mahalanobis distance of $\boldsymbol{x}$ from class centre $\boldsymbol{\mu}_c$ under precision matrix $\boldsymbol{\Sigma}_c^{-1}$ is:*

$$\delta^2(\boldsymbol{x}, c) = (\boldsymbol{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_c) \tag{2}$$

The unnormalized log-likelihood used by the classifier is:

$$\boxed{\log p(\boldsymbol{x} \mid c) = -\left[\delta^2(\boldsymbol{x},\, c)\right]^{\beta/2} = -\left[(\boldsymbol{x} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}_c^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_c)\right]^{\beta/2}} \tag{3}$$

Remark 1. *Equation (3) omits the normalizing constant (which does not depend on $\boldsymbol{x}$ for fixed β) because it cancels in the softmax posterior (Section 5). The normalizing constant involves the generalized gamma function and is not required for classification.*

4 Parameter Estimation

4.1 Class-Conditional Moments

Let $\mathcal{D}_c = \{\boldsymbol{x}_i : y_i = c\}$ be the training set for class c, with $N_c = |\mathcal{D}_c|$.

Mean vector.

$$\hat{\boldsymbol{\mu}}_c = \frac{1}{N_c} \sum_{\boldsymbol{x} \in \mathcal{D}_c} \boldsymbol{x} \tag{4}$$

Covariance matrix.

$$\hat{\boldsymbol{\Sigma}}_c = \frac{1}{N_c - 1} \sum_{\boldsymbol{x} \in \mathcal{D}_c} (\boldsymbol{x} - \hat{\boldsymbol{\mu}}_c)(\boldsymbol{x} - \hat{\boldsymbol{\mu}}_c)^\top + \varepsilon\, \mathbf{I} \tag{5}$$

where $\varepsilon = 10^{-6}$ is a Tikhonov regularization term added to ensure positive definiteness and numerical invertibility of $\hat{\boldsymbol{\Sigma}}_c$.

Class prior.

$$\hat{\pi}_c = \frac{N_c}{N_0 + N_1} \tag{6}$$

4.2 Shape Parameter

The shape parameter β is a hyperparameter set by the user (default $\beta = 1.5$). It is not estimated from data in the current implementation. Cross-validation over $\beta \in \{1.0, 1.5, 2.0\}$ is recommended to select the value that maximises validation AUC.

5 Posterior Probability and Classification

5.1 Posterior via Bayes' Rule

Applying Bayes' rule with the log-likelihoods from Eq. (3):

$$P(Y = c \mid \boldsymbol{x}) \propto p(\boldsymbol{x} \mid c)\, \pi_c \tag{7}$$

5.2 Numerically Stable Softmax

To avoid numerical overflow, we compute the log-posteriors and apply a log-sum-exp stabilization. Let:

$$\ell_c(\boldsymbol{x}) = \log p(\boldsymbol{x} \mid c) + \log \pi_c = -\delta^2(\boldsymbol{x}, c)^{\beta/2} + \log \hat{\pi}_c \tag{8}$$

Then the posterior probability of class 1 is:

$$\boxed{\hat{P}(Y = 1 \mid \boldsymbol{x}) = \frac{\exp(\ell_1(\boldsymbol{x}) - \ell_{\max}(\boldsymbol{x}))}{\sum_{c \in \{0,1\}} \exp(\ell_c(\boldsymbol{x}) - \ell_{\max}(\boldsymbol{x}))}} \tag{9}$$

where $\ell_{\max}(\boldsymbol{x}) = \max_c \ell_c(\boldsymbol{x})$.

5.3 Decision Rule

A pixel $\boldsymbol{x}$ is classified as positive if:

$$\hat{P}(Y = 1 \mid \boldsymbol{x}) \geq \tau \tag{10}$$

where τ is the decision threshold.

5.4 Optimal Threshold via Youden's J Statistic

The threshold τ^* is selected to maximise Youden's J statistic over the training data:

$$\tau^* = \arg\max_{\tau} \big[\mathrm{TPR}(\tau) - \mathrm{FPR}(\tau)\big] = \arg\max_{\tau} \; J(\tau) \tag{11}$$

where $\mathrm{TPR}(\tau) = \mathrm{sensitivity}(\tau)$ and $\mathrm{FPR}(\tau) = 1 - \mathrm{specificity}(\tau)$. This is equivalent to maximising the vertical distance from the ROC curve to the diagonal chance line, ensuring balanced sensitivity and specificity.

6 Band Importance Metrics

6.1 Permutation Importance

Let AUC_0 denote the baseline ROC-AUC on training data. The permutation importance of band b is:

$$I_b^{\mathrm{perm}} = \mathrm{AUC}_0 - \mathrm{AUC}\big(\hat{P}(Y = 1 \mid \boldsymbol{x}^{(\sim b)})\big) \tag{12}$$

where $\boldsymbol{x}^{(\sim b)}$ replaces band b with a random permutation of its values across all pixels, thereby destroying its relationship with the label.

6.2 1-D KL Divergence (per band)

The Kullback–Leibler divergence between the univariate Gaussian marginals of the two classes in band b is:

$$\mathrm{KL}\big(\mathcal{N}(\hat{\mu}_{1b}, \hat{\sigma}_{1b}^2) \,\|\, \mathcal{N}(\hat{\mu}_{0b}, \hat{\sigma}_{0b}^2)\big) = \log \frac{\hat{\sigma}_{0b}}{\hat{\sigma}_{1b}} + \frac{\hat{\sigma}_{1b}^2 + (\hat{\mu}_{1b} - \hat{\mu}_{0b})^2}{2\,\hat{\sigma}_{0b}^2} - \frac{1}{2} \tag{13}$$

where $\hat{\mu}_{cb}$ and $\hat{\sigma}_{cb}$ are the mean and standard deviation of band b within class c. This is a threshold-free measure of distributional separability in each band.

6.3 Pairwise 2-D KL Divergence

For a pair of bands (i, j), we fit bivariate Gaussians $\mathcal{N}(\hat{\boldsymbol{\mu}}_{c,ij}, \hat{\boldsymbol{\Sigma}}_{c,ij})$ to each class in the 2-D subspace, where $\hat{\boldsymbol{\Sigma}}_{c,ij}$ is the 2×2 submatrix of $\hat{\boldsymbol{\Sigma}}_c$ corresponding to bands i and j.

The closed-form KL divergence between two k-dimensional Gaussians is:

$$\mathrm{KL}(\mathcal{N}_1 \,\|\, \mathcal{N}_0) = \frac{1}{2}\left[\mathrm{tr}\Big(\hat{\boldsymbol{\Sigma}}_0^{-1}\hat{\boldsymbol{\Sigma}}_1\Big) + (\hat{\boldsymbol{\mu}}_0 - \hat{\boldsymbol{\mu}}_1)^\top \hat{\boldsymbol{\Sigma}}_0^{-1}(\hat{\boldsymbol{\mu}}_0 - \hat{\boldsymbol{\mu}}_1) - k + \ln \frac{|\hat{\boldsymbol{\Sigma}}_0|}{|\hat{\boldsymbol{\Sigma}}_1|}\right] \tag{14}$$

The **symmetric KL** (also called Jensen–Shannon surrogate) is:

$$\mathrm{KL}^{\mathrm{sym}}(i, j) = \mathrm{KL}(\mathcal{N}_1 \| \mathcal{N}_0) + \mathrm{KL}(\mathcal{N}_0 \| \mathcal{N}_1) \tag{15}$$

Interaction gain. The *interaction gain* of band pair (i, j) quantifies synergy beyond the sum of individual-band contributions:

$$G(i, j) = \mathrm{KL}_{\mathrm{2D}}^{\mathrm{sym}}(i, j) - \tfrac{1}{2}\big[\mathrm{KL}_{\mathrm{1D}}^{\mathrm{sym}}(i) + \mathrm{KL}_{\mathrm{1D}}^{\mathrm{sym}}(j)\big] \tag{16}$$

$G(i, j) > 0$ indicates synergistic bands (jointly more discriminative than their parts); $G(i, j) < 0$ indicates redundancy.

6.4 Mutual Information

Mutual information between band b and the label Y is:

$$I(X_b; Y) = H(Y) - H(Y \mid X_b) = \sum_c \int p(x, c) \ln \frac{p(x, c)}{p(x)\,p(c)}\, dx \tag{17}$$

In practice this is estimated using the k-nearest-neighbour entropy estimator of Kraskov et al. (2004), which makes no parametric distributional assumption and therefore captures non-linear dependencies missed by the Gaussian KL metric. High MI relative to KL rank for a given band indicates the presence of non-Gaussian class structure in that band.

7 Greedy KL-Based Band Selection (`--KL`)

When the `--KL` flag is specified, the tool performs greedy forward band selection to find the subset $\mathcal{S}^* \subseteq \{1, \ldots, B\}$ that maximises the joint symmetric KL divergence between the two classes.

Definition 2 (Joint symmetric KL for a band subset)**.** *For a subset $\mathcal{S}$ of bands, let $\mathcal{N}(\hat{\boldsymbol{\mu}}_{c,\mathcal{S}}, \hat{\boldsymbol{\Sigma}}_{c,\mathcal{S}})$ be the multivariate Gaussian fitted to class c in the restricted feature space. The objective is:*

$$J(\mathcal{S}) = \mathrm{KL}(\mathcal{N}_{1,\mathcal{S}} \,\|\, \mathcal{N}_{0,\mathcal{S}}) + \mathrm{KL}(\mathcal{N}_{0,\mathcal{S}} \,\|\, \mathcal{N}_{1,\mathcal{S}}) \tag{18}$$

Algorithm 1 Greedy Forward KL Band Selection

Input: $X \in \mathbb{R}^{N \times B}$, labels y, band names
Compute $J(\{b\})$ for each $b \in \{1, \ldots, B\}$
$\mathcal{S} \leftarrow \{\arg\max_b J(\{b\})\}$
$J^* \leftarrow J(\mathcal{S})$
$\mathcal{R} \leftarrow \{1, \ldots, B\} \setminus \mathcal{S}$
while $\mathcal{R} \neq \emptyset$ **do**
  $b^* \leftarrow \arg\max_{b \in \mathcal{R}} J(\mathcal{S} \cup \{b\})$
  if $J(\mathcal{S} \cup \{b^*\}) > J^*$ **then**
    $\mathcal{S} \leftarrow \mathcal{S} \cup \{b^*\}$
    $J^* \leftarrow J(\mathcal{S} \cup \{b^*\})$
    $\mathcal{R} \leftarrow \mathcal{R} \setminus \{b^*\}$
  else
    break ▷ No further improvement
  end if
end while
Output: $\mathcal{S}$

Remark 2. *This algorithm is greedy and does not guarantee a globally optimal subset. Exact optimization over all 2^B subsets is NP-hard for large B. The greedy approach has complexity $O(B^2)$ in the number of KL evaluations.*

8 Band-Name Validation in `--apply` Mode

During `--apply`, the classifier must locate the bands it was trained on within a potentially differently-ordered (or differently-populated) raster. The procedure is:

1. Load the serialized model. Extract the ordered list of band names $\mathcal{B}_{\text{model}} = [b_1, b_2, \ldots, b_K]$ stored at training time.
2. Read the target raster. Extract its band-name list $\mathcal{B}_{\text{raster}} = [r_1, r_2, \ldots, r_M]$.
3. For each $b_k \in \mathcal{B}_{\text{model}}$, find its index $\iota_k = \{j : r_j = b_k\}$ in the new raster. If any $b_k \notin \mathcal{B}_{\text{raster}}$, the program exits with an informative error listing the missing bands.
4. Construct the reordered feature matrix $X_{\text{apply}} = [X_{\iota_1}, X_{\iota_2}, \ldots, X_{\iota_K}]$ such that column k of X_{apply} corresponds to band b_k as expected by the model.

This guarantees that the model's covariance matrices and mean vectors — estimated in a fixed band order — are applied to the correct physical bands regardless of how the target raster was constructed.

9 Diagnostic Visualizations

The following figures are produced in `--fit` mode.

Figure	Description
Classification map	Binary prediction alongside ground-truth mask.
Probability map	Pixel-level posterior $\hat{P}(Y = 1 \mid \boldsymbol{x})$, hot colormap.
Mahalanobis map	Distance $\delta(\boldsymbol{x}, 1)$ from positive class centroid.
Correlation matrix	Pearson correlation among selected bands.
ROC curve	True positive vs. false positive rate; AUC annotated.
Confusion matrix	TP / FP / TN / FN counts.
Permutation importance	AUC drop when each band is randomly shuffled.
Mahalanobis distribution	Per-class histogram of $\delta(\boldsymbol{x}, 1)$ scores.
1-D marginals	Per-band histograms with class Gaussian curves, decision boundary, and KL annotation.
KL bar chart	Per-band $\mathrm{KL}(\mathcal{N}_1 \| \mathcal{N}_0)$ ranked.
MI bar chart	Mutual information per band (non-parametric).
MI vs KL scatter	Flags bands with non-Gaussian structure.
2-D interaction matrix	Upper: scatter + ellipses; Lower: positive-class fraction.
Pairwise KL heatmap	Both $\mathrm{KL}(+\|-)$ and $\mathrm{KL}(-\|+)$ for all band pairs.
Interaction gain	Band synergy / redundancy matrix (Eq. 16).

10 1-D Marginal Projection and Decision Boundary

The 1-D marginal of a multivariate Gaussian $\mathcal{N}(\hat{\boldsymbol{\mu}}_c, \hat{\boldsymbol{\Sigma}}_c)$ onto band b is:

$$X_b \mid Y = c \;\sim\; \mathcal{N}\big(\hat{\mu}_{cb},\ \hat{\sigma}^2_{cb}\big) \quad \text{where} \quad \hat{\sigma}^2_{cb} = [\hat{\boldsymbol{\Sigma}}_c]_{bb} \tag{19}$$

The **per-band decision boundary** is the value x^* at which the positive and negative class Gaussian densities are equal:

$$\mathcal{N}(x^* \mid \hat{\mu}_{1b}, \hat{\sigma}^2_{1b}) = \mathcal{N}(x^* \mid \hat{\mu}_{0b}, \hat{\sigma}^2_{0b}) \tag{20}$$

When $\hat{\sigma}_{1b} \neq \hat{\sigma}_{0b}$, this equation is quadratic in x and may yield zero, one, or two real solutions. In the implementation, the boundary is approximated numerically as:

$$x^* \approx \arg\min_{x \in \mathcal{X}_b} |\mathcal{N}(x \mid \hat{\mu}_{1b}, \hat{\sigma}_{1b}) - \mathcal{N}(x \mid \hat{\mu}_{0b}, \hat{\sigma}_{0b})| \tag{21}$$

where $\mathcal{X}_b$ is a dense grid over the 1st–99th percentile range of band b.

11 Model Serialization

11.1 Binary Format (.pkl)

The full model object is serialized using Python's `pickle` module. The stored object contains:

- β, `best_threshold`, `selected_bands`;
- `band_names`: ordered list of all band names seen during training;
- `params`: dict with keys $\{0, 1\}$, each containing $\hat{\boldsymbol{\mu}}_c$, $\hat{\boldsymbol{\Sigma}}_c$, and $\hat{\boldsymbol{\Sigma}}_c^{-1}$;
- `priors`: $\hat{\pi}_0$, $\hat{\pi}_1$.

11.2 Human-Readable Format (.txt)

A plain-text summary is written alongside the `.pkl` file, containing: all scalar parameters, selected band names, per-class mean vectors, diagonal variances, and full covariance matrices in array notation.

12 GDAL Raster I/O

All raster input and output is handled via the GDAL Python bindings (`osgeo.gdal`).

Reading. Each raster band is read as a flat float32 array; no-data values (as declared in the raster metadata) are replaced with `NaN`. Band descriptions (set via `band.GetDescription()`) are used as band names; bands without descriptions are named `band_1`, `band_2`, etc. Pixels where any band is non-finite are excluded from model fitting.

Writing (–apply). Two GeoTIFF files are produced:

- `<model>_prediction.tif`: Float32, posterior probability $\hat{P}(Y = 1 \mid \boldsymbol{x})$, LZW-compressed, tiled;
- `<model>_classified.tif`: Byte (0/1), binary classification, LZW-compressed, tiled.

Both files inherit the geotransform and projection of the input raster.

13 Command-Line Interface

```
python gg_classifier.py --fit  --image scene.tif --mask mask.tif  --model run1
python gg_classifier.py --fit  --KL --image scene.tif --mask mask.tif --model run1
python gg_classifier.py --apply --image new_scene.tif --model run1
```

Flag	Description
`--fit`	Fit a new model. Requires –image and –mask.
`--apply`	Apply a saved model. Requires –image and –model.
`--image`	Path to GDAL-readable raster.
`--mask`	Path to single-band 0/1 training mask (–fit only).
`--model`	File stem for .pkl and .txt serialization.
`--beta`	Shape parameter β (default 1.5).
`--bands`	Whitespace-separated band names to load (default: all).
`--KL`	Enable greedy KL band selection (–fit only).
`--out_dir`	Output directory for result rasters (default: current).

14 References

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